

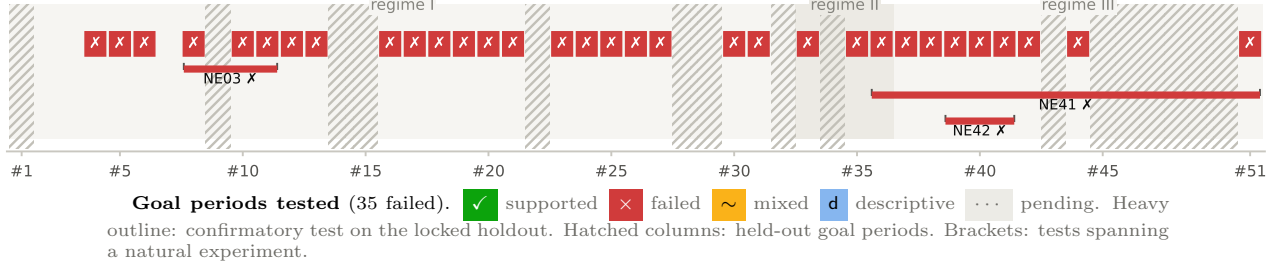
H45: Context homeostasis: each agent keeps its context near a set point

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 2/5

Bottom line. The room share of an agent's context is not regulated: it follows room traffic as passive accumulation predicts, and the scaffold's resets and windows, not the agents, bound it.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH166 claims that each agent holds the room-message share of its context near a set point. The agent reads less per message when the room floods ($k^{-0.6}$ dilution as the homeostat), and overshoots, then relaxes, after an erasure. H45 computes the share per model call from DQ1's context ledger and measured prompt tokens, on 32 non-holdout periods (#4–#51). Four native tests add NE41's 21k forced erasures, the NE42 merge, #51's roster growth and NE03's chat fetch limit. Synthetic controller, passive and scaffold-capped agents on real call schedules come first. **The claim is refuted.** The share follows inflow as passive accumulation predicts: the regulation index is ≤ 0.22 in 32/32 periods, pooled 0.02. Floods move set points at 1.06 (NE42) and 0.90 (#51) times the passive slope. After an erasure, agents talk *less* (0.74) and reply more per talk; replies per call do not change. The holdout script is written, not run.

PHYSICAL MODEL

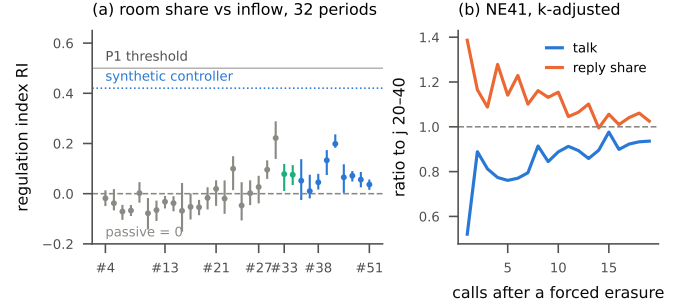
Models 04 and 05 frame a set-point controller against passive accumulation. In one context segment between resets, room content R_j (calibrated tokens of the new room items) and own content W_j (the rest of the prompt P_j) accumulate. The order parameter is the share $s_j = R_j/P_j$, and the control parameter is λ , the room tokens per call. A controller would raise W or reset sooner when the share is high, and engage more when it is low. ε is the elasticity of the segment share to λ , and η_w that of own content.

$$s_j = \frac{R_j}{R_j + W_j}, \quad R_j = \sum_{i \leq j} (a k_i + b c_i + e o_i)$$

$$\text{passive: } s_j = \frac{\lambda_j}{W_j + \lambda_j}, \quad W_j \text{ independent of } \lambda$$

$$\varepsilon = \left. \frac{\partial \ln \bar{s}}{\partial \ln \lambda} \right|_j = (1 - \bar{s})(1 - \eta_w), \quad \text{RI} = 1 - \frac{\varepsilon}{1 - \bar{s}} = \eta_w$$

$$P(\text{reply}) = 1 - \exp(-\theta_{\text{agent} \cdot \text{day}} k^{1-\beta})$$



Observables. (a) Regulation index per period (day-bootstrap CI; regime I gray, II aqua, III blue). Passive accumulation gives 0. The synthetic controller on the same schedules gives 0.42; the prediction needs ≥ 0.5 . Every period sits near passive. (b) NE41: after a forced erasure (41-call cap), the k-adjusted talk propensity dips about 26% and recovers over about ten calls. The share of talks that reply to pending messages rises. Replies per call stay flat (0.92).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Agents do not protect their context from a flooding room, and they do not “refill” it after an erasure. Room traffic per call, the agent's own tool output and the scaffold's resets set the share of a context window spent on chat. To cut chat in context, change the room (size, merges) or the scaffold (cadence, caps); agents do not ration it. A room merge doubles the share. The 200-event cap almost never binds (0.05% of calls). An erasure costs about ten calls of reduced talk, not a catch-up burst. Engagement behaves like context competition: replies fall as the agent's own work fills the context. This gives no lever on the agents themselves. It is a design rule: the scaffold decides context composition.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 $RI \geq 0.5$ in $\geq 2/3$	0/32; CI upper < 0.5 in 32/32; pooled 0.02	failed
P3 share band (CV ratio ≤ 0.5)	0.84–1.32, median 0.99	failed
P2 reset sooner when flooded	3/9 regime III; #51 -0.04 (controller -0.10)	failed (weak)
P4 dilution $\beta \in [0.4, 0.9]$	21/32; pooled 0.82 (III 0.98); H18 ρ 0.71	narrowly failed
P5 $\gamma_R < 0$, $\gamma_W > 0$ (pooled III)	$\gamma_R +0.08$, $\gamma_W -0.10$ (42k talks)	failed
P6 post-erasure over-shoot	0/9; pooled talk 0.74 [0.69, 0.80]	reversed
P7 lab sets the set point	lab η^2 0.14, p 0.41	failed
P8/P9 no truncation; cap rare	P doubles in-segment; cap 0.05%	passed
NE42 / #51 leak ≤ 0.3	1.06 [0.95, 1.11] / 0.90 [0.77, 0.99]	failed
NE03 chat ceiling ap-pears	flat before and after (8–10k to-kens)	failed

GOAL PERIODS IN DETAIL

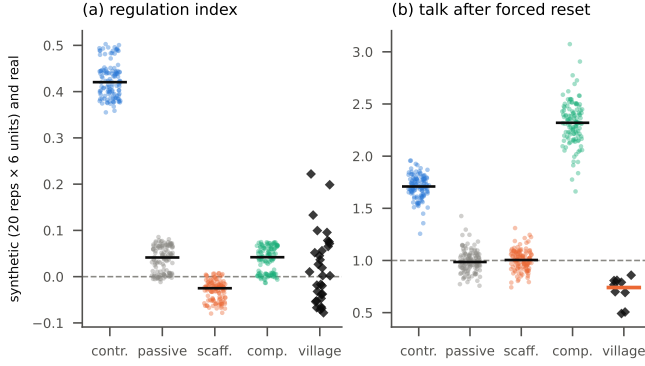
G04 $\times$ failed (RI -0.02 [-0.05, 0.01])
G05 $\times$ failed (RI -0.04 [-0.06, 0.01])
G06 $\times$ failed (RI -0.07 [-0.10, -0.05])
G08 $\times$ failed (RI -0.07 [-0.08, -0.05])
G10 $\times$ failed (RI 0.00 [-0.03, 0.04])
G11 $\times$ failed (RI -0.08 [-0.14, -0.02])
G12 $\times$ failed (RI -0.07 [-0.10, -0.03])
G13 $\times$ failed (RI -0.03 [-0.05, -0.01])
G16 $\times$ failed (RI -0.04 [-0.07, -0.01])
G17 $\times$ failed (RI -0.07 [-0.15, 0.04])

+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Synthetic validation on real schedules (6 units, 20 replicates; dots), with the village (diamonds: per-period estimates; orange bar: pooled NE41). At village counts, the regulation index separates a controller from passive, scaffold and competition agents. The village falls in the passive column. After erasures it talks less than any synthetic class.

CAVEATS

Room content R is a lower bound: `search_history` results are not counted, and the reset snapshot is ≈ 0 . Shared error in R pulls ε toward passive, so a weak controller ($RI < 0.15$) is hard to see. Dense prompt tokens exist only for Anthropic and Google. One DQ2 reply parent per message steepens β . The engagement-on-context test is not identified per period, and task phase confounds it. The goal confounds NE42. The holdout is not used.

ROUND 2

- **Essence.** whether agents manage their information intake at all, or the scaffold does it for them.
- **H45-R1** Context competition (HH158): does reply engagement fall with own-content fill once DQ3 behavior states control task phase?
- **H45-R2** Cadence as the agents' only intake lever: do call intervals shorten when the room's per-minute rate rises?
- **H45-R3** What fills the post-erasure talk dip?
- **H45-R4** Run `confirm.py` on the holdout (#45–#50, 51m, NE22) after the reuse disclosures.

H45-context-homeostasis · generated 2026-10-04 12:01 by infra/summaries/build_summaries.py · card hypotheses/H45-context-homeostasis/README.md · ratings: H45 agent (suggested) (page 2)